

Project Title

Healthcare Readmission Prediction using Machine Learning

Project Overview

This project focuses on predicting hospital readmission of diabetic patients using machine learning techniques. The goal is to identify whether a patient is likely to be readmitted within 30 days, enabling better healthcare decision-making and resource optimization.

Tech Stack

- Python
 - Pandas, NumPy
 - Scikit-learn
 - XGBoost
 - SMOTE (Imbalanced data handling)
 - Matplotlib, Seaborn
 - SHAP (Model Explainability)
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Dataset

- Dataset: Diabetes hospital records
 - Includes patient demographics, medical history, and hospital visit details
 - Target variable: **Readmission (<30 days)**
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Key Steps

1. Exploratory Data Analysis (EDA)

- Analyzed age distribution, gender split, and readmission rates
- Identified missing values and class imbalance
- Visualized trends using graphs

2. Data Preprocessing

- Removed irrelevant columns
- Handled missing values
- Converted categorical variables using Label Encoding
- Created binary target variable

3. Handling Imbalanced Data

- Applied **SMOTE** to balance the dataset

4. Model Building

Trained and compared multiple models:

- Logistic Regression
- Random Forest
- XGBoost

5. Evaluation

- Used **ROC-AUC score** for performance comparison
- Selected best-performing model
- Visualized model accuracy

6. Model Explainability

- Used **SHAP** to interpret feature importance and model decisions

Results

- XGBoost performed best among all models
- Improved prediction accuracy after applying SMOTE
- Key features influencing readmission were identified using SHAP

Conclusion

The project successfully predicts hospital readmissions using machine learning. This can help healthcare providers take preventive measures and improve patient care.
